
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Although reinforcement learning (RL) has recently produced substantial improve-
2 ments in the reasoning capabilities of language models, the operative training
3 dynamics through which RL reshapes LLM behavior remain insufficiently charac-
4 terized. In this work, we offer an explanatory account of the RL training process
5 by combining systematic empirical dissection with a rigorous theoretical model.
6 Through reasoning-pattern-level and token-level analyses conducted throughout
7 training, we show that, even as distinct reasoning patterns retain comparatively
8 stable success rates, RL concentrates its optimization on a sparse subset of conse-
9 quential tokens and thereby reorganizes the distribution over reasoning patterns in
10 ways that materially affect model performance. Motivated by these observations,
11 we formulate a theoretical framework for two representative reward mechanisms:
12 verifiable rewards (RLVR) and the model’s internal feedback (RLIF). For RLVR,
13 we analyze two special regimes, one in which models readily converge toward opti-
14 mal reasoning strategies and another in which optimization becomes substantially
15 more difficult, thereby revealing the decisive role played by the base model’s initial
16 reasoning quality in determining convergence behavior. For RLIF, we character-
17 ize how internally generated rewards can yield early performance improvements
18 while also admitting subsequent degradation under continued training. Extensive
19 experiments substantiate these theoretical claims, advancing both the mechanistic
20 understanding and the practical deployment of RL for language-model reasoning
21 enhancement.

22 1 Introduction

23 Recent state-of-the-art reasoning models, including Gemini2.5 [8], Qwen3 [40], and DeepSeek-
24 R1 [14], have exhibited exceptional performance on complex logical tasks spanning mathematics
25 [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL), increasingly recognized as
26 a central post-training mechanism underlying these advances, appears capable of elevating model
27 competence beyond the level attained by conventional supervised procedures alone.

28 The empirical success of RL has consequently stimulated a growing body of work devoted to
29 uncovering its mechanisms in LLMs. By comparing pass@k performance, Yue et al. [42] show that
30 models post-trained with RL struggle to surpass their base counterparts, suggesting that RL may
31 not elicit fundamentally novel reasoning patterns. From an entropy-oriented perspective, Cui et al.
32 [9], Zhang et al. [45] theoretically prove that RL-based methods can reduce policy entropy, while Cui
33 et al. [9] further establishes an empirical connection between model performance and policy entropy.
34 Complementarily, Wang et al. [33], Huan et al. [17] demonstrate that RL primarily optimizes a sparse
35 subset of critical tokens. Despite these convergent attempts to interpret RL mechanisms, however, the
36 fine-grained dynamics of the RL training process remain only partially understood, both empirically
37 and theoretically.

To clarify these dynamics, we first conduct a systematic analysis of RL training at both the reasoning-pattern and token levels. Rather than merely inspecting static before-and-after comparisons, we examine ranking shifts throughout training and use both LLM-based and rule-based procedures to extract, classify, and track reasoning patterns in model responses, together with their associated success rates and distributional evolution. Relative to prior analyses [17, 42], our experiments provide sharper evidence that RL chiefly optimizes a sparse set of high-leverage tokens, thereby *reshaping reasoning pattern distributions* and altering model performance through distributional reallocation rather than wholesale invention of new reasoning capabilities. Moreover, we observe that the intrinsic success rates of individual patterns remain comparatively stable, an empirical regularity that motivates a mathematical account of the RL training process.

Building on these findings, we develop a theoretical framework that conceptualizes reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: first, the model selects a reasoning pattern conditioned on the question, represented by $\pi(r|q)$; subsequently, it generates an answer conditioned on both the question and the selected pattern, represented by $\pi(a|r, q)$. Within this framework, we analyze RL training dynamics for two representative reward types: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we show convergence toward the reasoning pattern r^* with the highest success rate, in close correspondence with our empirical observations, and we further distinguish two convergence regimes: strong initial reasoning quality induces rapid convergence to optimal patterns, whereas weaker base models encounter optimization difficulty during an entanglement stage. For RLIF, we explain why internal rewards can initially improve performance, while also showing that the same training procedure may ultimately converge to a state whose performance is worse than that of the base model. Additional experiments corroborate these theoretical conclusions.

The main contributions of this paper are summarized as follows:

- We provide a systematic reasoning-pattern-level and token-level analysis of the RL training process. By jointly examining next-token prediction ranking shifts and LLM-based or rule-based reasoning-pattern decompositions, we offer more fine-grained evidence than previous work [17, 42] regarding how RL training evolves. Our experiments show that RL predominantly optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* and changing model performance, while the intrinsic success rates of individual patterns remain relatively stable.
- We introduce a formal two-stage mathematical framework that represents reasoning as $q \rightarrow r \rightarrow a$ (question-reason-answer) and use it to analyze the training dynamics of two typical RL-based approaches. For RLVR, we prove convergence to the reasoning pattern with the highest success rate and characterize two regimes: rapid convergence for strong base models and entanglement-stage optimization challenges for weaker models. For RLIF, we provide a theoretical justification for early-stage performance gains and explain why continued training may nevertheless yield performance below that of the base model.
- We validate the theoretical analysis through additional case studies and simulations. By connecting empirical observations with a tractable formal model, our framework offers practical insight into how RL-based LLM post-training can be understood and improved.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has become an influential instrument for improving large language models (LLMs), particularly in preference alignment [25, 48] and in the solution of mathematically or programmatically demanding tasks [30, 20, 22]. A central component of this paradigm is the reward model, which has traditionally depended on human-annotated datasets [26, 25, 2] and substantial training effort. More recently, however, the field has shifted toward reward signals that are easier to obtain and verify, as exemplified by reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. In parallel, the empirical success of RL has motivated a proliferation of new optimization algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The success of RL in LLM post-training has also generated extensive work on theoretical characterization and mechanistic interpretation. Early investigations primarily pursued high-level questions, such as how rewards

should be designed [28, 18, 38] and how novel training algorithms should be constructed [37, 11, 21]. As the field has moved toward simpler reward structures such as RLVR and RLIF, researchers have increasingly sought to explain the fundamental sources of their effectiveness from multiple perspectives. Shao et al. [29] argue that spurious rewards can improve reasoning by unlocking latent skills acquired during pretraining, whereas Agarwal et al. [4] attribute performance gains to entropy minimization. Gandhi et al. [13] identify particular cognitive behaviors, including backtracking, that contribute to stronger reasoning. Additional evidence from [46] indicates that RL fine-tuning amplifies pretrained behaviors and induces convergence toward dominant output formats, a phenomenon further analyzed through entropy collapse by Wang et al. [34], Cui et al. [10]. In contrast to these prior perspectives, our study analyzes RL training dynamics through the lens of reasoning-pattern selection and supports this interpretation with a mathematical framework and formal training-dynamic analysis.

3 Preliminaries

Reinforcement Learning for LLMs Let π_θ denote a language model with parameters θ , interpreted as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ produces an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$, and the corresponding optimization objective can be written as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} denotes the base reference policy, and β is a hyperparameter that regulates the KL penalty so as to constrain excessive deviation between π_θ and π_{ref} .

The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ may be instantiated in several ways, including as a trained reward model [25] or as a rule-based scoring function [30]. In this paper, we focus on two representative choices: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

Under RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the generated answer $\mathbf{y}$ matches the ground-truth answer associated with question $\mathbf{x}$. A standard binary reward can therefore be specified as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

In contrast, RLIF [47, 4] constructs rewards exclusively from intrinsic, model-derived signals. Following Zhao et al. [47], we take the RLIF reward to be the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution,

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

Our goal is therefore to characterize, through both empirical and theoretical analysis, how RLIF and RLVR induce their respective training dynamics.

4 Experimental Exploration for RL Training

To illuminate the RL training process, we conduct a systematic empirical investigation across training runs induced by different reward mechanisms. We begin with a macroscopic account of the training trajectories, which shows that RLVR produces steady improvement whereas RLIF can behave unstably and may even degrade performance. We then decompose these trajectories at the reasoning-pattern level, showing that the effectiveness of RLVR derives from its progressive allocation of probability mass toward patterns with higher success rates, while RLIF does not reliably specialize in such patterns. Finally, we analyze the token-level mechanism underlying these distributional dynamics and find that they are driven by rank changes affecting only a surprisingly small fraction of tokens. Compared with previous works [17, 42], our experiments provide more direct and fine-grained evidence that RL primarily optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* in ways that affect model performance.

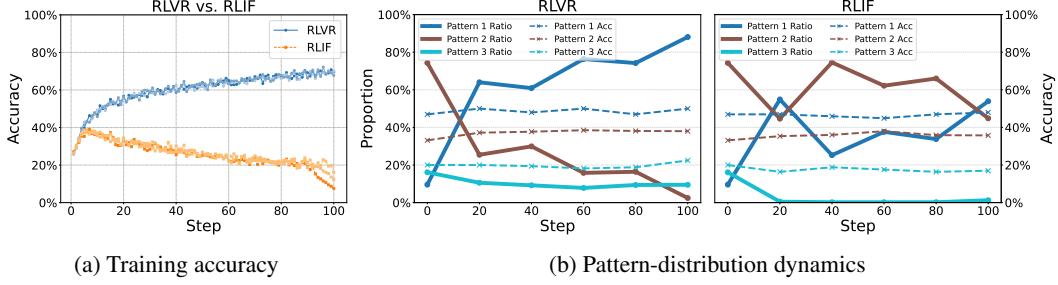


Figure 1: *(a)* Training procedures for RLVR and RLIF, measured by performance on the MATH dataset. The principal observation is that RLVR yields stable gains, whereas RLIF first increases and then decreases model performance. For each training paradigm, we conduct multiple experimental rounds by varying the random seed. *(b)* Reasoning-pattern-level comparison. The solid line denotes the response-level proportion of a given pattern among all extracted patterns (left vertical axis), whereas the dotted line denotes the accuracy associated with that pattern (right vertical axis). During RLVR, the model increasingly adopts higher-accuracy patterns and suppresses lower-accuracy patterns; RLIF, by contrast, exhibits unstable training dynamics. For visual clarity, patterns are sorted and named from high to low according to their corresponding accuracy.

4.1 Training Procedure Overview for RLVR and RLIF

We first examine the complete training procedures associated with RLVR and RLIF. To ensure a controlled comparison, we use Qwen2.5-3B [39] as the base model and train it on the MATH dataset [16], while holding all remaining experimental settings fixed across the two algorithms.

As Figure 1a illustrates, the two methods follow markedly different performance trajectories. RLVR induces a comparatively stable training process in which model performance improves continuously and gradually converges. By contrast, although RLIF can improve performance during the early stages of training, additional optimization may precipitate a sharp decline, in some cases producing a model whose performance falls below that of the original base model. These divergences motivate a more systematic and fine-grained empirical analysis of the underlying RL training dynamics.

4.2 Reasoning-pattern Level Analysis

To explain the performance disparities between RLVR and RLIF, we conduct a fine-grained analysis at the level of reasoning patterns. We operationalize these patterns by first collecting responses from the base model and then using GPT-4o [19] to cluster them into distinct categories according to lexical cues, keywords, and logical structure, while also producing a description for each category. These classifications subsequently allow us to track both the distribution over reasoning patterns and the success rate associated with each pattern throughout RL training.

More specifically, we analyze how reasoning-pattern distributions and pattern-specific accuracies evolve under both RLVR and RLIF. Figure 1b reports this analysis for responses to Number Theory tasks drawn from the MATH dataset [16], while Appendix A provides additional results across other models and datasets. The resulting evidence yields three central findings:

- RLVR-trained models consistently reallocate probability mass toward reasoning patterns with higher success rates, thereby explaining the steady improvement in overall accuracy under RLVR.
- The reasoning-pattern distribution induced by RLIF exhibits unstable training dynamics and does not reliably specialize in the more effective patterns.
- For both methods, the success rate of each individual reasoning pattern remains comparatively stable throughout the training process.

This analysis indicates that the performance gap between RLVR and RLIF arises from their distinct reasoning-pattern dynamics. We therefore conduct additional experiments to identify the lower-level mechanisms that drive these distributional changes during RL training.

4.3 Token-level Analysis

To probe the mechanisms underlying the reasoning-pattern dynamics observed during RL training, and following the motivation of Huan et al. [17], Yue et al. [42], we further conduct a token-level analysis. We first sample responses from the base model for questions drawn from GSM8K [6], MATH [16], and AIME [7]. For these identical responses, we then compare the corresponding token ranks under the base model and the RL-enhanced model, where a token’s rank denotes the position of its probability among all vocabulary tokens conditioned on the preceding context. As summarized in Table 1, individual-token ranks remain largely stable: in most cases, fewer than 10% of tokens undergo a rank shift. This result indicates that RL selectively modifies the probabilities of a sparse set of critical tokens while leaving the majority of the reasoning trajectory effectively unchanged.

Taken together, these experiments provide concise but compelling evidence about RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning-pattern distributions in ways that affect model performance, while the intrinsic success rates of individual patterns remain relatively stable during training.*

These empirical findings both deepen the mechanistic understanding of RL training dynamics and motivate the mathematical framework used in the subsequent theoretical analysis.

Table 1: Ratio of token-rank changes after RL.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

5 Theoretical Characterization of RL Training Dynamics

Moving beyond empirical observation, we develop a theoretical characterization of the RL training process. We first construct a framework, motivated by the experimental findings, that abstracts model reasoning into a two-step procedure: the selection of a reasoning pattern followed by answer deduction, with RL primarily acting on the former component. Within this framework, we analyze the behaviors induced by different reward mechanisms. For RLVR, we characterize the training dynamics and isolate two special regimes with distinct optimization profiles. For RLIF, we explain how internal rewards can initially improve model performance while also clarifying why the same procedure may ultimately yield models whose performance is worse than that of the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize reasoning as follows. Given a question q , the model first selects a reasoning pattern r from a candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and then generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that selection. The model chooses pattern r_i with probability $p(r_i|q)$, and each pattern is associated with its own success rate. Under this abstraction, the RLIF optimization objective in Eq. 3.3 can be reformulated as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set, and let $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ denote the set of admissible output sequences. Conditioned on a question q , the language model autoregressively induces a distribution over output sequences $y \in \mathcal{Y}$:

$$(\text{general policy}) \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

Here, $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ , and the model predicts the l -th token from the preceding context $y_{<l}$. For $z = f_\theta(x, y_{<l})$, the next token is sampled according to $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$.

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

Because practical language models are highly complex, deriving optimization guarantees that directly explain RL training dynamics is technically difficult [3, 23]. Following prior work [27, 24, 10, 45], we therefore study a *tabular policy* parameterization, which can be interpreted as a special case of Eq. 5.2 in which each output receives its own trainable logit for the corresponding last token; namely, for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$(\text{tabular policy}) \quad \pi_{\theta}(\mathbf{y}_l | \mathbf{x}, \mathbf{y}_{<l}) = \pi_{\theta}(\mathbf{y}_l | \mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

where $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{y}_{l-1}$.

Optimization for RL. To analyze training dynamics, we combine the tabular policy in Eq.5.3 with the RL objective $\phi_{\text{RL}}(\theta)$ in Eq.3.1. In the small-learning-rate limit, the policy-gradient update is represented by the following gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

Our experiments indicate that the success rates of individual reasoning patterns often remain stable during training. We attribute this regularity to architectural constraints in the model: optimizing the mapping from questions to reasoning patterns is substantially easier than modifying the downstream mapping from reasoning patterns to final answers. We formalize this observation with the following assumption:

Assumption 5.1. *For each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$, the success rate of producing the correct answer $\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^* | \mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

For the remainder of this section, we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^* | \mathbf{q}, \mathbf{r})$ for the fixed success rate of pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective in Eq.3.1 under the general autoregressive policy in Eq.5.2; we then analyze the training dynamics of RLVR and RLIF using the tabular policy in Eq. 5.3.

5.2 The Optimal Policy for RL

Our empirical results show that RLVR steadily improves model performance by incentivizing reasoning patterns with higher success rates, whereas RLIF produces a more unstable improvement profile. We first explain these differences by characterizing the optimal policy associated with the RLVR and RLIF objectives:

Proposition 5.2. *Suppose the RL objective in Eq 3.1 is maximized using the general autoregressive policy in Eq 5.2, and suppose Assumption 5.1 holds. Then the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r} | \mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- For RLIF, $R(\mathbf{r})$ corresponds to the confidence of the final-answer distribution induced by reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a} | \mathbf{q}, \mathbf{r}))$.

Proposition 5.2 characterizes the optimal solutions for both RLVR and RLIF. Provided that π_{opt} lies within the general autoregressive policy parameterization (as it does for both LLMs and the tabular policies analyzed below), RLVR increases the probability of a reasoning pattern $\mathbf{r}$ whenever its probability of deriving the correct answer, $p^*(\mathbf{r})$, is sufficiently large that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. By contrast, $R_{\text{RLIF}}(\mathbf{r})$ depends only on the confidence of the final-answer distribution and therefore does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Consequently, whereas the RLVR optimal policy can consistently favor improved performance, the RLIF optimal policy provides no guarantee of accuracy improvement over the base model.

Moreover, because β is typically small in practice (e.g., 0.001) [26, 5, 44], the limit $\beta \rightarrow 0$ yields:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the policy becomes deterministic and always selects the reasoning pattern $\mathbf{r}$ with the largest $R(\mathbf{r})$, irrespective of the reasoning-pattern distribution induced by the initialized reference model. Nevertheless, because the optimization landscape is non-convex, the mechanisms by which RLVR reliably discovers high-reward patterns, and by which RLIF can exhibit early performance gains, remain nontrivial. We therefore analyze the training dynamics of both methods under the tabular policy in Eq. 5.3, obtaining a more detailed account of the RL training process.

5.3 Training Dynamic Analysis for RLVR

Our experiments indicate that RLVR improvements depend strongly on the capacity of the base model. We focus on the probability assigned to the *optimal reasoning pattern*, namely the pattern $\mathbf{r}^*$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$. Through gradient-flow analysis, we identify two distinct regimes in the RLVR dynamics. In the first regime, the probability of the optimal reasoning pattern, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$, increases monotonically until it converges to 1. In the second regime, the model initially enters an entanglement stage in which a suboptimal reasoning pattern $\mathbf{r}' \neq \mathbf{r}^*$ impedes optimization toward the optimal pattern. After this entanglement stage, the dynamics transition into the first regime and ultimately converge to $\mathbf{r}^*$.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 considers the favorable case in which the base reference model is sufficiently strong that its overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under this condition, RLVR efficiently guides the model toward the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$, thereby attaining high overall accuracy close to the success rate of that optimal pattern.

In practice, however, RLVR optimization can be difficult in some settings [44, 36], especially when the base reference model is less capable. We therefore analyze a second regime in which the model initially experiences an entanglement stage, and the *suboptimal reasoning pattern with the second-highest success rate*, defined as $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, slows optimization toward the optimal reasoning pattern:

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(\text{Regime 2}) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 isolates a special regime in which only the optimal and suboptimal reasoning patterns have success rates above the overall accuracy. We adopt this setting for analytical tractability, and we expect the argument to extend to more general cases in which *at least* two reasoning patterns exceed the average success rate, although such extensions would require a more involved analysis.

The central implication of Theorem 5.4 is that RLVR may require T_0 steps before the overall success rate surpasses that of the suboptimal reasoning pattern, thereby entering the favorable regime of Theorem 5.3 in which the optimal reasoning pattern has sufficiently large selection probability. Although Theorem 5.3 shows that the convergence time T_1 scales polynomially in $1/\epsilon$, the threshold T_0 in Theorem 5.4 may grow super-exponentially with $\gamma_{\pi_{\text{ref}}}$ —the ratio between the total success rate through $\mathcal{R}/\{r'\}$ and the success rate through r^* . Thus, when the base model assigns extremely small probability to the optimal reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ can be very large, making T_0 prohibitively large as well. In such cases, the model may require an extremely long training period before it assigns the optimal reasoning pattern a reasonably large probability; we refer to this prolonged phase as the entanglement stage.

Additional experiments and case studies illustrating these two regimes are provided in Sec 6.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 shows that RLIF, because it considers only the confidence of the final-answer distribution, does not distinguish the correct answer a^* from incorrect alternatives. This observation explains why RLIF may eventually degrade performance relative to the base model, consistent with our empirical findings. At the same time, prior studies have shown that RLIF can improve model performance without external rewards Agarwal et al. [4], Zhao et al. [47]. We therefore provide a theoretical explanation for why RLIF can enhance the performance of a well-trained LLM during the initial stage of training.

We begin with a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer a^* has the highest probability among all possible answers, i.e., $\arg \max_{a \in \mathcal{A}} \sum_{r \in \mathcal{R}} \pi_{\text{base}}(a|q, r) \pi_{\text{base}}(r|q) = a^*$.*

Because majority voting [35] is an effective means of improving model performance, well-trained LLMs typically assign higher aggregate probability to the correct answer a^* than to any incorrect answer; consequently, Assumption 5.5 is plausible for modern LLMs. Under this constraint, we consider the case in which $|\mathcal{A}| = 2$, the success rate of each reasoning path is drawn from a uniform distribution, and the reasoning-pattern selection distribution is high-entropy². We then analyze the overall accuracy dynamics at $t = 0$, obtaining the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(r|q) = \frac{1}{|\mathcal{R}|}$ for all $r \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(r) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. *The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.*
2. *With probability $p = 0.5$, that $\arg \max_r R_{\text{RLIF}}(r) = \arg \min_r p^*(r)$.*

The first part of Theorem 5.6 shows that overall accuracy increases at initialization, thereby providing a theoretical explanation for the early performance gains observed under RLIF. As training proceeds, however, Proposition 5.2 implies that, when $\beta = 0$, the policy converges deterministically to the reasoning pattern r that maximizes $R_{\text{RLIF}}(r)$. The second part of Theorem 5.6 shows that, with probability 0.5, this maximizing pattern is also the pattern with the minimum success rate: $\arg \max_r R_{\text{RLIF}}(r) = \arg \min_r p^*(r)$. Equivalently, there is a 50% chance that the model converges to the least accurate reasoning path. Thus, although RLIF initially improves performance because the accuracy derivative is positive, continued training can drive the model toward states that favor very low-accuracy reasoning paths, ultimately producing performance below that of the base model.

6 Case Studies and Numerical Simulations

To further validate and interpret the theoretical findings, we present case studies and numerical training simulations that instantiate the preceding theorems. For RLVR, Fig. 2 (upper) illustrates the two training regimes characterized in Theorems 5.3 and 5.4. For RLIF, Fig. 2 (bottom) reports

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(r|q) = 1/|\mathcal{R}|$ for all $r \in \mathcal{R}$.

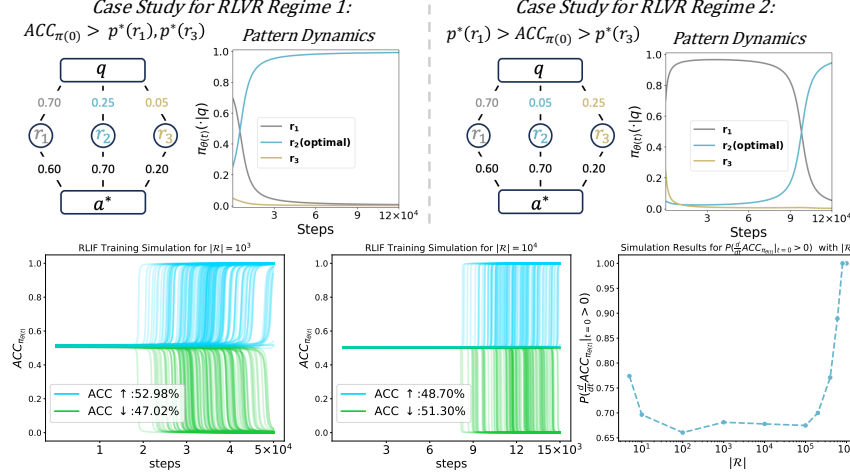


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels illustrate two distinct RLVR training regimes: rapid convergence (left) and the entanglement phase (right). The lower panels report, across varying $|\mathcal{R}|$, the probabilities of convergence to different states (left) and the probability of initial performance gains under RLIF (right).

multiple training simulations in which each reasoning path’s success rate is sampled from a uniform distribution, thereby validating our analysis of RLIF behavior both at the initial training steps and at convergence.

6.1 Case Studies for RLVR

Section 5.3 identifies two regimes in the RLVR training dynamics. In the first regime (Eq 5.7), the probability of selecting the optimal reasoning pattern, $\pi_{\theta}(r^*|q)$, increases monotonically until convergence. In the second regime (Eq 5.8), the model first enters an entanglement phase and requires a substantially longer training interval before transitioning to the first-regime dynamics and converging to the optimal reasoning pattern r^* . We examine both regimes as follows:

- As shown in Fig. 2 (upper left), when the reference model satisfies Eq 5.7, so that its overall accuracy exceeds the success rates of all non-optimal reasoning patterns, the model rapidly converges to the optimal reasoning pattern.
- In the second case, corresponding to Theorem 5.4, a suboptimal reasoning pattern r' has a higher success rate than the initial overall accuracy, causing the model to experience an entanglement phase. The transition period T_0 defined in Theorem 5.4 must elapse before the model enters the rapid optimization phase characteristic of the first case. Fig. 2 (upper right) shows a setting in which $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9) is large ($\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$). In this situation, the transition time T_0 , which scales with $\gamma_{\pi_{\text{ref}}}$, becomes prohibitively long and substantially delays convergence to the optimal reasoning pattern. The pattern dynamics in Fig. 2 accordingly exhibit the predicted entanglement stage and align with the theoretical result.

6.2 Numerical Simulations For RLIF

Section 5.4 shows that, under Assumption 5.5 and the base-model distribution specified in Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the probability of convergence to the least accurate reasoning path is 50%. Our simulations validate these claims. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$, confirming that the probability of convergence to an improved state ($\text{ACC} \uparrow$) is approximately equal to the probability of convergence to a degraded state ($\text{ACC} \downarrow$), as predicted by the second part of Theorem 5.6. In addition, the initial-step analysis in Fig. 2 (bottom right) shows that, for large $|\mathcal{R}|$, the probability of initial performance improvement approaches 1, supporting the first part of Theorem 5.6.

7 Conclusions and Limitations

This work analyzes reinforcement-learning dynamics in LLMs through a combination of empirical investigation and theoretical formalization. We develop mathematical analyses for two representative reward mechanisms, RLVR and RLIF, and substantiate the resulting claims through case studies and simulations. The analysis nevertheless has several limitations that warrant further investigation: the interpretability of LLM-identified reasoning patterns requires additional validation, the theoretical framework could be extended to accommodate more complex real-world reasoning scenarios, and the present analysis could be generalized beyond the specific assumptions imposed on the base model.

References

- [1] MAA. 2023. American mathematics competitions. https://artofproblemsolving.com/wiki/index.php/American_Mathematics_Competitions, 2023. Online.
- [2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [3] Alekh Agarwal, Sham M Kakade, Jason D Lee, and Gaurav Mahajan. On the theory of policy gradient methods: Optimality, approximation, and distribution shift. *Journal of Machine Learning Research*, 22(98):1–76, 2021.
- [4] Shivam Agarwal, Zimin Zhang, Lifan Yuan, Jiawei Han, and Hao Peng. The unreasonable effectiveness of entropy minimization in llm reasoning. *arXiv preprint arXiv:2505.15134*, 2025.
- [5] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- [6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [7] MAA Codeforces. American invitational mathematics examination-aime 2024, 2024.
- [8] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- [9] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, Zhiyuan Liu, Hao Peng, Lei Bai, Wanli Ouyang, Yu Cheng, Bowen Zhou, and Ning Ding. The Entropy Mechanism of Reinforcement Learning for Reasoning Language Models, May 2025.
- [10] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, et al. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025.
- [11] Nirjhar Das, Souradip Chakraborty, Aldo Pacchiano, and Sayak Ray Chowdhury. Active preference optimization for sample efficient rlhf. In *ICML 2024 Workshop on Theoretical Foundations of Foundation Models*, 2024.
- [12] Amir Dembo. *Large deviations techniques and applications*. Springer, 2009.
- [13] Kanishk Gandhi, Ayush Chakravarthy, Anikait Singh, Nathan Lile, and Noah D Goodman. Cognitive behaviors that enable self-improving reasoners, or, four habits of highly effective stars. *arXiv preprint arXiv:2503.01307*, 2025.

- [14] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [15] Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, et al. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. *arXiv preprint arXiv:2402.14008*, 2024.
- [16] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.
- [17] Maggie Huan, Yuetai Li, Tuney Zheng, Xiaoyu Xu, Seungone Kim, Minxin Du, Radha Pooven-dran, Graham Neubig, and Xiang Yue. Does Math Reasoning Improve General LLM Capabili-ties? Understanding Transferability of LLM Reasoning, July 2025.
- [18] Jiawei Huang, Bingcong Li, Christoph Dann, and Niao He. Can rlhf be more efficient with im-perfect reward models? a policy coverage perspective. In *ICLR 2025 Workshop on Bidirectional Human-AI Alignment*, 2025.
- [19] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- [20] Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, et al. Openai o1 system card. *arXiv preprint arXiv:2412.16720*, 2024.
- [21] Kaixuan Ji, Jiafan He, and Quanquan Gu. Reinforcement learning from human feedback with active queries. *arXiv preprint arXiv:2402.09401*, 2024.
- [22] Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, et al. Tulu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124*, 2024.
- [23] Gen Li, Yuting Wei, Yuejie Chi, Yuantao Gu, and Yuxin Chen. Softmax policy gradient methods can take exponential time to converge. In *Conference on Learning Theory*, pages 3107–3110. PMLR, 2021.
- [24] Jincheng Mei, Chenjun Xiao, Csaba Szepesvari, and Dale Schuurmans. On the global con-vergence rates of softmax policy gradient methods. In *International conference on machine learning*, pages 6820–6829. PMLR, 2020.
- [25] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- [26] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36:53728–53741, 2023.
- [27] Noam Razin, Zixuan Wang, Hubert Strauss, Stanley Wei, Jason D Lee, and Sanjeev Arora. What makes a reward model a good teacher? an optimization perspective. *arXiv preprint arXiv:2503.15477*, 2025.
- [28] Antoine Scheid, Etienne Boursier, Alain Durmus, Michael I Jordan, Pierre Ménard, Eric Moulines, and Michal Valko. Optimal design for reward modeling in rlhf. *arXiv preprint arXiv:2410.17055*, 2024.
- [29] Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, et al. Spurious rewards: Rethinking training signals in rlvr. *arXiv preprint arXiv:2506.10947*, 2025.

- [30] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Y Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [31] Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. *arXiv preprint arXiv: 2409.19256*, 2024.
- [32] Qwen Team. Qwq-32b: Embracing the power of reinforcement learning, March 2025. URL <https://qwenlm.github.io/blog/qwq-32b/>.
- [33] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive Effective Reinforcement Learning for LLM Reasoning, June 2025.
- [34] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025.
- [35] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- [36] Tian Xie, Zitian Gao, Qingnan Ren, Haoming Luo, Yuqian Hong, Bryan Dai, Joey Zhou, Kai Qiu, Zhirong Wu, and Chong Luo. Logic-rl: Unleashing llm reasoning with rule-based reinforcement learning. *arXiv preprint arXiv:2502.14768*, 2025.
- [37] Wei Xiong, Hanze Dong, Chenlu Ye, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, and Tong Zhang. Iterative preference learning from human feedback: Bridging theory and practice for rlhf under kl-constraint. In *International Conference on Machine Learning*, pages 54715–54754. PMLR, 2024.
- [38] Shusheng Xu, Wei Fu, Jiaxuan Gao, Wenjie Ye, Weilin Liu, Zhiyu Mei, Guangju Wang, Chao Yu, and Yi Wu. Is dpo superior to ppo for llm alignment? a comprehensive study. In *International Conference on Machine Learning*, pages 54983–54998. PMLR, 2024.
- [39] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [40] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [41] Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- [42] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.
- [43] Yu Yue, Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiaze Chen, Chengyi Wang, TianTian Fan, Zhengyin Du, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks. *arXiv preprint arXiv:2504.05118*, 2025.

- 504 [44] Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He.
505 Simplertl-zoo: Investigating and taming zero reinforcement learning for open base models in the
506 wild. *arXiv preprint arXiv:2503.18892*, 2025.
- 507 [45] Yanzhi Zhang, Zhaoxi Zhang, Haoxiang Guan, Yilin Cheng, Yitong Duan, Chen Wang, Yue
508 Wang, Shuxin Zheng, and Jiyan He. No Free Lunch: Rethinking Internal Feedback for LLM
509 Reasoning, June 2025.
- 510 [46] Rosie Zhao, Alexandru Meterez, Sham Kakade, Cengiz Pehlevan, Samy Jelassi, and Eran
511 Malach. Echo chamber: RL post-training amplifies behaviors learned in pretraining. *arXiv
512 preprint arXiv:2504.07912*, 2025.
- 513 [47] Xuandong Zhao, Zhewei Kang, Aosong Feng, Sergey Levine, and Dawn Song. Learning to
514 reason without external rewards. *arXiv preprint arXiv:2505.19590*, 2025.
- 515 [48] Banghua Zhu, Michael Jordan, and Jiantao Jiao. Principled reinforcement learning with human
516 feedback from pairwise or k-wise comparisons. In *International Conference on Machine
517 Learning*, pages 43037–43067. PMLR, 2023.
- 518 [49] Qihao Zhu, Daya Guo, Zhihong Shao, Dejian Yang, Peiyi Wang, Runxin Xu, Y Wu, Yukun
519 Li, Huazuo Gao, Shirong Ma, et al. Deepseek-coder-v2: Breaking the barrier of closed-source
520 models in code intelligence. *arXiv preprint arXiv:2406.11931*, 2024.

A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

In Section 4.2, we analyze reasoning-pattern dynamics using Number Theory tasks sampled from MATH. We additionally sample Geometry tasks for the same dynamic analysis, as shown in Figure 3. During RLVR, the model progressively adopts reasoning patterns with higher accuracy, whereas the pattern-selection behavior under RLIF remains unstable. This result is consistent with both the empirical observations and the theoretical conclusions reported in the main paper.

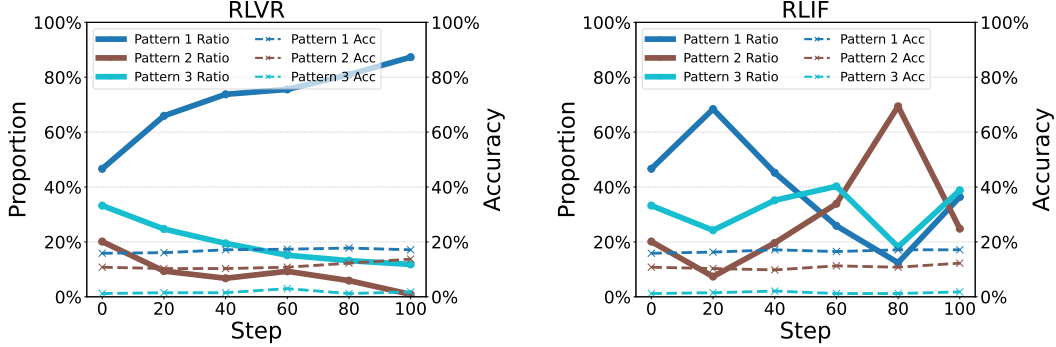


Figure 3: Dynamics of pattern distributions and corresponding accuracies for RL-enhanced models on Geometry tasks. Under RLVR, the model tends to select the highest-accuracy pattern (Pattern 1), whereas RLIF exhibits unstable selection behavior.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To assess whether our findings transfer across models, we conduct supplementary experiments that further validate the empirical insights and theoretical conclusions (Figure 4). In these experiments, Qwen-2.5-7B-Instruct [39] serves as the base model and is compared with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. The evaluation spans multiple mathematical domains, including number theory, geometry, algebra, calculus, and counting and probability, using challenging problems sampled from the MATH dataset [16]; we additionally include complex mathematical problems from AMC23 [1]. The results consistently reinforce our earlier findings: RLVR enhancement increases the prevalence of high-accuracy reasoning patterns while reducing the frequency of less effective patterns. Detailed task-specific analyses are provided in Appendix B.3.

Across these task-level reasoning-pattern analyses, the patterns with the highest success rates consistently become more prevalent after RLVR enhancement, reinforcing the conclusions of Section 4 and Section 5. Notably, for tasks such as Algebra and AMC23-19 in Figure 4, the success rates of individual reasoning patterns remain stable, which both agrees with our earlier observations and supplies empirical support for Assumption 5.1 in the theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To further examine the theory at larger scale, we conduct experiments on QwQ-32B [32]. This model is derived from Qwen-2.5-32B-Instruct [39] and substantially improves reasoning ability through RLVR. We evaluate the reasoning-pattern shifts of these two models on four tasks: number theory and geometry tasks from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The results show that most reasoning patterns produced by QwQ-32B correspond to the most accurate patterns identified for Qwen-2.5-32B, in strong agreement with our theoretical results. We do not report QwQ pattern accuracies because the model is evaluated through an API, which prevents extraction of the CoT data needed for the base model to continue generating answers. Consequently, we cannot obtain the accuracy of the reasoning patterns themselves, as distinct from the model’s overall accuracy. Further details appear in the “Accuracy Analysis” discussion in Appendix B.2.

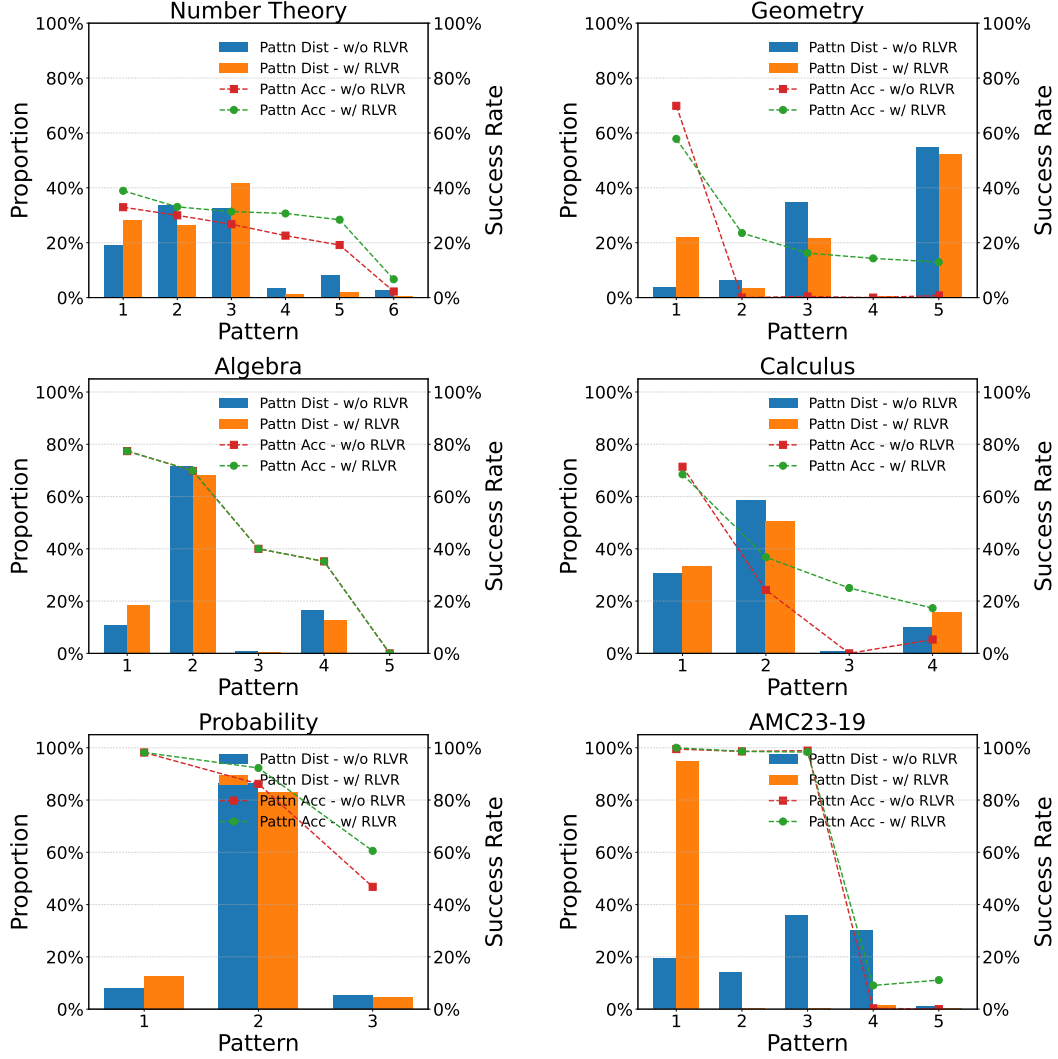
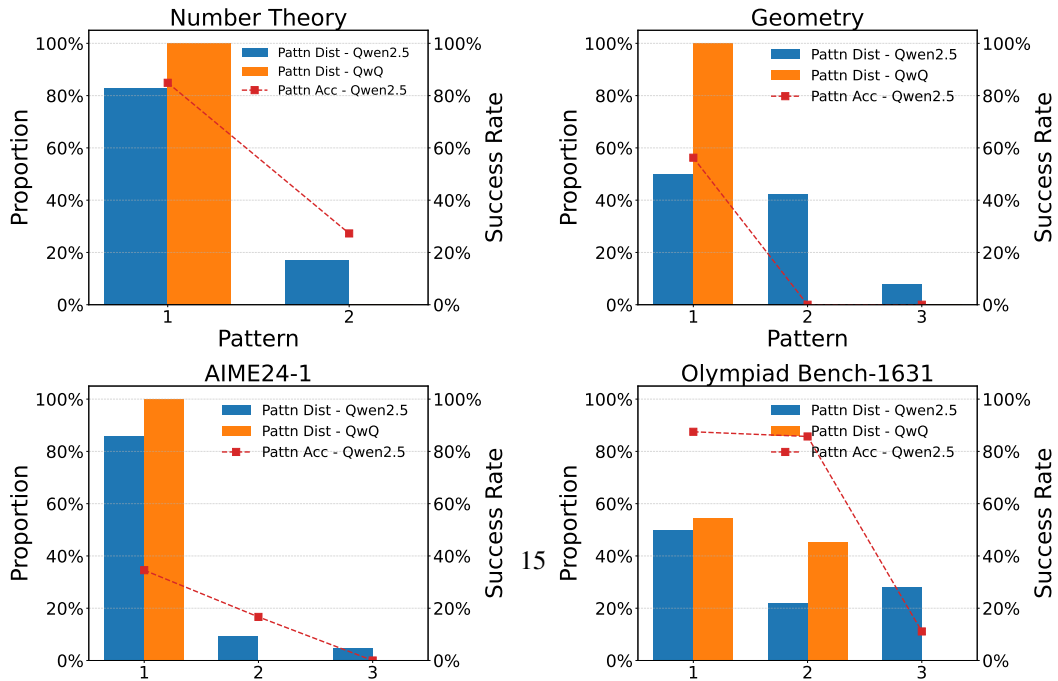


Figure 4: Evaluation results for reasoning patterns and corresponding success rates with and without RLVR enhancement across additional tasks, aligning with our empirical insights and theoretical conclusions. Bars (Pattern Dist) denote the proportion of a given pattern among all extracted patterns, while dots (Pattern Acc) denote the accuracy or success rate associated with that pattern.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We use Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. The experiments adopt the verl [31] framework, with all parameter settings matched to the MATH example provided by verl. For RLIF, the only modification is the reward-calculation procedure. By varying the random seed, we obtain three distinct training curves for each method.

Token-level Analysis: This experiment uses Qwen2.5-3B together with the RL-enhanced models described above. For GSM8K and MATH, we randomly sample 400 questions from each test set, while for AIME24 we use the complete dataset. For each question, we first sample an answer from the base model; we then concatenate that answer with the question and provide the resulting sequence to the RL-enhanced model as a new prompt. Using the interface in the OpenAI library, we compute the probability of each token in the prompt and its rank among all possible tokens at the corresponding position. By comparing token ranks before and after RL, we calculate the proportion of tokens whose ranks change.

Reasoning-pattern Level Analysis: We conduct two categories of experiments for reasoning-pattern analysis. First, using previously trained Qwen2.5-3B and RL-enhanced Qwen2.5-3B models, we study the dynamics of the pattern distribution during training and evaluate the model every 20 steps. Second, to assess whether the conclusions generalize across base models and larger model scales, we evaluate Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. For these externally available models, we can examine only the before-and-after changes in output pattern distributions, rather than the full pattern-distribution dynamics during training.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We randomly sample questions from multiple task types, such as Geometry and Algebra, and from different difficulty levels, including Level 4 and Level 5, in datasets such as MATH. For each question, the model samples 1024 answers in the pattern-distribution dynamic analysis; for experiments that compare only the before-and-after distributional change induced by RL, the number of sampled answers ranges from 64 to 1024 depending on the task category. These responses are then used for subsequent pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of base-model responses and use the GPT-4o [19] API to summarize pattern categories from those samples. The complete prompt used for reasoning-pattern extraction appears in Appendix D.1. To improve the determinism and consistency of the extraction procedure, we set the GPT-4o API temperature to 0.

Responses Classification: We again employ GPT-4o, with temperature set to 0, to classify all responses according to the identified pattern categories. The complete response-classification prompt is provided in Appendix D.2.

Accuracy Analysis: To isolate the effect of pattern distributions on accuracy, we construct new prompts by concatenating the original prompt with partial responses from the base model, under the assumption that these partial responses sufficiently instantiate specific reasoning patterns. This construction fixes the reasoning process, making it easier to determine whether changes in model accuracy arise from changes in the reasoning-pattern distribution rather than from changes in the reasoning process itself. Concretely, we remove the sentence containing the final answer from the base-model response, append the preceding sentence to the end of the question, and input the resulting prompt to the RL-enhanced model, which then continues generation. We treat the accuracy of the predicted answer as the accuracy associated with the corresponding reasoning pattern.

B.3 Details for Each Task

We next detail the experimental procedures for each task, including task descriptions and examples of reasoning patterns (Table 2). Example task prompts are provided in Appendix D.3.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from } 243) + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Examples of reasoning patterns across tasks.

Number Theory: This task presents models with problems involving coin distribution across multiple bags. Initially, bags contain equal numbers of coins. After receiving additional coins and redistributing them equally, the total must exceed a specified value while maintaining equal distribution. Models must determine the minimum initial coin count per bag. We derived this task from a level 5 MATH dataset problem (id: test/number_theory/1055) [16], creating 32 variants by adjusting parameters like bag count and coin totals as our evaluation dataset. We employ one-shot prompting with a simple, unrelated example to guide answer formatting using "boxed" notation without influencing reasoning approaches. For evaluation, we first choose 20 questions from our evaluation dataset with 4 responses each for reasoning pattern extraction. For each question we sample 64 responses. We then categorize all responses by the extracted reasoning patterns, allowing us to compare changes in the distribution of models' reasoning patterns before and after RL.

Geometry: For the geometry task, we provide the model with the lengths of two sides of a scalene triangle and ask how many distinct integer-centimeter lengths are possible for the third side. This task is derived from a Level 4 Geometry problem in MATH (id: test/geometry/1046). The evaluation follows the same pipeline as Number Theory: we construct a synthetic dataset containing 32 questions, sample 20 questions and select 4 responses per question from the base model (Qwen2.5-7b-Instruct) for reasoning-pattern extraction, and then compare responses from models with and without RLVR enhancement on all questions in the evaluation dataset, using 64 responses per question for pattern analysis.

Algebra: The algebra task is a can-recycling problem. Starting from a specified number of old cans, and given that n old cans can be recycled into one new can, the model must determine the total

number of cans that can ultimately be produced. The main difficulty is that each iterative recycling step may leave extra cans that cannot be evenly divided at that stage but may later be combined with newly produced cans for further recycling. The task template is derived from a Level 4 Algebra problem in MATH (id: test/algebra/2768). For this and the subsequent supplementary tasks, we use the same settings as in the preceding tasks, including dataset size and sampling counts.

Calculus: The calculus task is deliberately simple and direct: we present a trigonometric equation in one variable, specify the variable’s domain, and ask the model to determine the number of solutions. Although the problem is not highly complex, it probes the model’s basic competence on calculus-adjacent reasoning. The prototype is a Level 4 Precalculus problem in MATH (id: test/precalculus/1140).

Counting and Probability: We also evaluate the model on a counting-and-probability task involving white and black balls. Two balls are sampled at random, and the prompt provides the probability that one drawn ball is black and the other is white, together with the number of balls of one color; the model must compute the minimum possible number of balls of the other color. This probability task is derived from a Level 4 Counting and Probability problem in MATH (id: test/counting_and_probability/79).

AMC23 Question: We use Question 19 from AMC23 [1]. For reasoning-pattern extraction, we randomly sample 48 responses, and for reasoning-pattern analysis we then sample 1024 responses.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We conduct experiments on four tasks. Because API-based testing imposes access limitations, we randomly select one question per task as input, sample 64 responses from each of the two models, and directly perform reasoning-pattern extraction and classification on these answers. For the two MATH tasks, we use Question 1055 from Number Theory and Question 1046 from Geometry. For the remaining tasks, we use Problem 1 from AIME24 and Problem 1631 from OlympiadBench. Examples of the resulting patterns are shown in Table 2.

C Deferred Proofs

This appendix supplies the proofs of our main theoretical results: Proposition 5.2, Theorem 5.3, Theorem 5.4, and Theorem 5.6.

C.1 Proof of Proposition 5.2

Proof of Proposition 5.2. The proof follows the techniques of [26]. Recall that the RL optimization objective is

$$\phi_{\text{RL}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under the framework introduced in Section 5.1, the policy reasons by first sampling $\mathbf{r}_i \in \mathcal{R}$ according to $\pi_{\boldsymbol{\theta}}(\mathbf{r}_i | \mathbf{q})$ and then generating the final answer $\mathbf{a} \in \mathcal{A}$ according to $\pi_{\boldsymbol{\theta}}(\mathbf{a} | \mathbf{r}_i)$. The reward $\phi_{\text{RL}}(\boldsymbol{\theta})$ can therefore be written as

$$\phi_{\text{RL}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{y} \sim \pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r}) r_\phi([\mathbf{r}, \mathbf{a}], q)$, then

$$\begin{aligned} \phi_{\text{RL}}(\boldsymbol{\theta}) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q}), \mathbf{a} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[\ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] + \beta \ln Z. \end{aligned}$$

The third equality follows from Assumption 5.1, and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})$ is the partition constant ensuring that

$$\pi^*(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}),$$

is a valid probability distribution satisfying $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r} \mid \mathbf{q}) = 1$. Since Z does not depend on $\mathbf{r}$.

Consequently, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y} \mid \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} \mid \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] - \beta \ln Z. \end{aligned}$$

By the nonnegativity and minimization properties of KL divergence, the optimal policy for the KL-constrained reward maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

Specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$; for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

Since the second term is identical for all $\mathbf{r} \in |\mathcal{R}|$, the RLIF optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) &= \frac{1}{Z} \exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \\ &= \frac{\exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i)) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})} \\ &= \frac{\exp \left(\frac{1}{\beta} (R'(\mathbf{r}) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})} \end{aligned}$$

Thus, $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r}))$. This completes the proof of the theorem. $\square$

670 C.2 Dynamics of Tabular Policy

671 Consider the RL objective in Eq 3.1 under the tabular policy in Eq 5.3, with Assumption 5.1 imposed.
 672 The gradient dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt}\theta(t) &= \nabla\phi_{\text{RL}}(\theta(t)) \\ &= \nabla\mathbb{E}_{\mathbf{r}\sim\pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r}\in\mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})\nabla\pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

673 Here, $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ denotes the column of θ corresponding to $\mathbf{q}$, and
 674 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ denotes the KL-regularized reward, so

$$\begin{aligned}\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j\in\mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla\text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j\in\mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

675 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r}\in\mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, this
 676 gives

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j\in\mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t}$$

677 C.3 Proof of Theorem 5.3

678 To prove Theorem 5.3, we first establish that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then
 679 derive a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5) and finally obtain the bound on t stated in Theorem
 680 5.3.

681 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

682 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j\in\mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j\in\mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q})\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

683 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

684 Since $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$. For the optimal reasoning pattern
685 $\mathbf{r}^*$, let $\mathbf{r}'$ denote the suboptimal reasoning pattern with the second-highest success rate, $\mathbf{r}' =$
686 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$. We then have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

687 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
688 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

689 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

690 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

691 This concludes the proof of the theorem. □

692 C.4 Proof of Theorem 5.4

693 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$ (Eq C.6), then we analyze
694 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

695 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 696 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

697 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

698 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

699 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 700 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

701 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

702 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

703 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

704 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

705 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

706 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

707 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

708 This concludes the proof of the theorem. $\square$

709 C.5 Proof of Theorem 5.6

710 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
711 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

712 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 713 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

714 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 715 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 716 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 717 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

718 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 719 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 720 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 721 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

722 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 723 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

724 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 725 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

726 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

727 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 728 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 729 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 730 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 731 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 732 $1 - 6\delta$.
- 733 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 734 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

735 This concludes the proof of the second statement.

736

□

737 D Full Prompts

738 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

739

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

740

741 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

742

743 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

744

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

745

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

746

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

747

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

748

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

749

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

750

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

751

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

752

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a s -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

753

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

754